THROUGH THE TENSOR

TPUs or Tensor Processing Units may hold the key to accelerate the future of machine learning big time

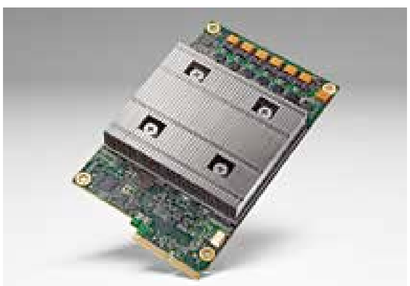
Jayesh Shinde
jayesh@digit.in

Earlier this year, at two different Google events, the company's CEO spoke at length of the search-engine giant's significant advances in the field of machine learning, deep neural networks and best-in-class cloud services. Amidst all the bells and whistles, one particular announcement made analysts sit up and take notice. Google's revelation of TPU, a piece of silicon the company had designed and fabricated to power its machine learning-tuned data centres. Why? Because Google thinks the TPU's implementation is pivotal to accelerate machine learning by leaps and bounds.

What's a TPU?

A Tensor Processing Unit is a proprietary chip designed by Google for use specifically for machine learning applications inside its TensorFlow framework – which is an open source software library for machine learning applications. Instead of a CPU or GPU which are designed to undertake a whole host of instruction sets for different software-side tasks and execute them with various degrees of efficiency, a TPU is an ASIC or application-specific integrated circuits chip that's only designed and programmed for machine learning instructions allowing it to be extremely efficient – certainly more efficient than a CPU or GPU undertaking the same task, according to claims made by Google. The company has only released the following image of what a TPU looks like, and other essential hardware and performance specifications are still a closely guarded secret at Google.

Yes, it's difficult to decipher a whole lot from this image of a TPU, as released by Google, but this is all we have for now. Wonder what lies beneath that heatsink, and what does the silicon die actually look like. Google has hinted at the fact that the TPU has the same footprint as a hard drive – we



Behold, for this is Google's own TPU!

don't know whether they meant it in terms of the unit's dimensions or power efficiency or both.

The fundamental way in which a TPU differs from a standard CPU (from Intel or AMD) or GPU (from NVIDIA or AMD) is its computational efficiency. Where a CPU and GPU is designed on a 64-bit architecture, experts are of the opinion that Google's TPU will most likely be running on 8-bit or 16-bit architecture which is far less precise in comparison. The advantage of sticking to an 8-bit or 16-bit cycle allows more logical circuits to be physically installed on the chip, allowing TPUs to calculate more operations per second and thereby be more efficient (in general and precise terms) than CPUs or GPUs.

Through TPUs and advancements in machine learning, Google claims it has advanced its hardware competency by up to three generations (or cycles) of Moore's Law. This is critical to give the company and its services a decisive edge over competition, as machine learning is used by over 100 product teams in Google – from Natural Language Processing (voice search), Smart Reply in Inbox, Street View in Maps, and a whole lot more.